

UNIVERSITY OF NAMIBIA

School of Engineering and the Built Environment | JEDS Campus
Computer Programming I (I3691CP) | Semester 1, 2026

INDIVIDUAL CONTRIBUTION REPORT

Load IQ — Ball Mill Stress Analysis App

Field	Details
Full Name	Chomore K
Student Number	225126591
Assigned Role	Documentation Lead
Group Name	Load IQ
Engineering Domain	Metallurgical Engineering
Submission Date	13 June 2026

0. Project Conception

In the early stages of the project, before the group had settled on an app idea, I took the initiative to visit lecturers’ offices across the department in search of a real engineering problem that our app could solve. Rather than relying on generic or fabricated scenarios, I wanted our project to be grounded in genuine, ongoing engineering work at the university. During these visits, I met with Dr Ilunga from the Department of Metallurgy at UNAM. Dr Ilunga introduced me to a ball mill project he is actively working on at the university. Through our discussion, I identified a specific and practically relevant engineering problem: calculating the stress at the support structure of the ball mill relative to the load inside it. This problem was both technically meaningful and well-suited to a mobile application format.

I brought the idea back to the group, where it was discussed and formally approved as the foundation of our project. This is the engineering problem that the Load IQ app is built around: providing a tool to compute and assess support stress in a ball mill based on variable load inputs. Video evidence of the site visit and the discussion with Dr Ilunga is available in the blog as supporting documentation.

1. Overview of My Contribution

As Documentation Lead, I was responsible for compiling and submitting all written deliverables on time. This included the System Requirements Specification (SRS), User Manual, and supporting documents for the Load IQ project.

2. Detailed Contributions

2.1 System Requirements Specification (SRS)

- Led the writing and compilation of the SRS document across Weeks 5 to 8
- Authored Section 1: Introduction, project title, scope, and problem statement
- Authored Section 2: System Overview including target users and technology overview
- Compiled Section 3: 15 Functional Requirements (FR-001 to FR-015) with priorities and Firebase service mapping
- Authored Section 4: Non-Functional Requirements covering performance, security, usability, and reliability
- Compiled Section 5: Use Case Diagrams and narratives for UC-01 to UC-08
- Authored Section 6: Engineering Threshold Logic with material yield strength reference table
- Authored Section 7: UI/UX Design Plan and screen inventory
- Authored Section 8: System Architecture and recommended folder structure
- Authored Section 9: Project Timeline and future improvements

2.2 User Manual

- Authored the Load IQ User Manual covering installation, registration, login, and all app features
- Documented the stress calculation procedure step by step
- Documented the result interpretation guide with colour-coded classification table
- Documented the history and delete functionality
- Included the material threshold reference table and troubleshooting section

2.3 Submission Management

- Submitted the SRS to the UNAM portal by the 25 April 2026 deadline
- Committed the SRS to the GitHub /docs folder
- Committed the User Manual to the GitHub /docs folder
- Ensured all documents followed the correct naming convention specified in the project brief

3. GitHub Commit Summary

All commits are visible in the public repository at:

<https://github.com/heleenkanmbunga/UNAM-I3691CP-LoadIQ-BallMillStressApp>

Commit Message	Phase
docs: Add SRS document to /docs folder	Phase 2
docs: Add User Manual to /docs folder	Phase 4
docs: Add Individual Contribution Report	Phase 4
docs: Add project scope and problem statement to SRS	Phase 2
docs: Add functional requirements section to SRS	Phase 2
docs: Add use case diagrams to SRS	Phase 2
docs: Add non-functional requirements to SRS	Phase 2
docs: Add README with group members and app description	Phase 0

4. Lines of Code / Work Contributed

My primary contributions were written documentation. I authored the SRS (submitted via UNAM portal and GitHub), the User Manual, and this Individual Contribution Report. I also committed all documentation to the GitHub repository.

5. Declaration

I declare that the contributions described in this report are my own original work. I have not misrepresented the work of other group members, and I have not violated UNAM's Academic Integrity Policy.

Name: Chomore K

Student Number: 225126591

Date: 13 June 2026

Load IQ | Individual Contribution Report | I3691CP | Semester 1, 2026